Henry Hsu

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SUMMARY

Software Engineering Intern at Google and former GenAI Team Lead, shipping production GenAI platforms and agentic developer infrastructure. Grew a B2B platform's client base by 120% and user base to 20K via Agentic AI Systems, Advanced RAG, and backend infrastructure, earning two major industry awards. Now pursuing an M.C.S. at Rice University.

WORK EXPERIENCE

Software Engineering Intern — Jun. 2026 – Aug. 2026

Google — *Taipei, Taiwan*

- **Integration Test Infrastructure:** Built **3** test infrastructures from a **zero** baseline (Java, Python), onboarding **400+** integration test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Agentic Failure Triage:** Developed an LLM agent that triages scheduled integration test runs and surfaces root-cause summaries directly on issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Device Reproduction:** Extended the agent to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

Generative AI Team Lead — Dec. 2024 – Aug. 2025

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Building AI Agents:** Upgraded chatbots into AI agents by integrating memory systems, tool APIs, and MCP Client, resulting in **120%** partner growth (CTBC Bank, MSI, HPE, iGroup) and user expansion from **3K** to **20K** users while cutting LLM token usage by **67%+**
- **Contributing to Open Source:** Contributed **14** merged pull requests to the **LlamaIndex** open-source project (**15K+ stars**), fixing bugs and adding features in integrations with AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, MCP Client, and AI Agent Workflow
- **Testing Framework:** Introduced pytest testing framework for unit and E2E coverage to prevent production regressions, achieving **67%** test coverage from **zero** baseline, which reduced production hotfixes by **90%** initially and sustaining a **50%** reduction long term
- **Technical Roadmap & Leadership:** Defined GenAI product roadmap for a 10-person startup; led feasibility evaluation and end-to-end delivery of production features including Agentic RAG, Artifact Generation, Tagging & Permission System, and Information Retrieval

Generative AI Intern — Sep. 2024 – Dec. 2024

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Pipeline Scaling:** Refactored with async framework and dual-database synchronization (RDB/Vector DB), achieving **3.5x** faster parsing and scaling from **3M** to **20M+** text chunks while **doubling** indexing performance
- **Reranker Integration:** Integrated Cohere and BGE open-source rerankers as a customer-selectable retrieval feature, improving RAG precision on large documents by **30% Precision@5** and **7% Precision@10**

EDUCATION

Rice University - Top 20 U.S. University — Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00) — *Houston, TX*

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University — Sep. 2020 – Jun. 2024

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) — *Hsinchu, Taiwan*

Minor: Computer Science (Domain GPA: 4.13/4.30)

PROJECTS

Prompt-Aware Image Token Pruning for Multimodal Inference | *PyTorch, LLaVA, CLIP ViT, Knowledge Distillation* **Feb. 2026 – May 2026**

- **Attention Distillation:** Taught a **1.3M-parameter** scorer to imitate where LLaVA-1.5-7B actually looks, dropping the patches it ignores before the projector – **10**× less attention compute for the frozen LLM, and **150**× cheaper than a CLIP-GradCAM pass
- **Accuracy & Latency:** Kept a quarter of the image tokens and still answered TextVQA about as well as the full model, **12.3** points clear of random pruning, while deciding what to keep in **0.43 ms** an image – cheap enough to be free at inference
- **Empirical Diagnostic:** Showed why the training-free shortcut fails: across three VQA benchmarks, what CLIP finds salient barely tracks what LLaVA attends to, which is what made learning the scorer worth the trouble

AWARDS

Presidential Hackathon Winner (2024, Top 5 Nationally) - Urban noise agent: structured-output LLM pipeline and tool calling

23rd Golden Peak Award (2025) - Outstanding Commercial Product, MaiAgent AI Platform

SKILLS

Languages: Python, Java, C/C++, TypeScript | **GenAI:** LlamaIndex, LangChain, LangGraph, MCP | **Fullstack:** Django, FastAPI, React
Databases: PostgreSQL, MongoDB, Milvus, Elasticsearch | **Tools:** nvim, git, Docker, Shell Script, Nginx
Data Science: numpy, pandas, scikit-learn, PyTorch, JAX | **AWS:** Bedrock, EMR, EC2, S3, RDS, ElastiCache, SageMaker